

Sentinel-1A SAR and sentinel-2A MSI data fusion for urban ecosystem service mapping



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ABSTRACT

The two main objectives of this study are to evaluate the potential use and synergetic effects of ESA Sentinel-1A C-band SAR and Sentinel-2A MSI data for classification and mapping of ecologically important urban and peri-urban space and to introduce spatial characteristics into ecosystem service analyses based on remotely sensed data. Image resolutions between 5 m and 20 m provided by the Sentinel satellites introduce a new relevant spatial scale in-between high and medium resolution data at which not only urban areas but also their important hinterlands can be effectively and efficiently mapped. Sentinel-1/2 data fusion facilitates both the capture of ecologically relevant details while at the same time also enabling large-scale urban analyses that draw surrounding regions into consideration. The combined use of Sentinel-1A SAR in Interferometric Wide Swath mode and simulated Sentinel-2A MSI (APEX) data is being evaluated in a classification of the Zürich metropolitan area, Switzerland. The SAR image was terrain-corrected, speckle-filtered and co-registered to the simulated Sentinel-2 image. After radiometric and spatial resampling, the fused image stack was segmented and classified by SVM. After post-classification, landscape elements were investigated in terms of spatial characteristics and topological relations that are believed to influence ecosystem service supply and demand, i.e. area, contiguity, perimeter-to-area ratio and distance. Based on the classification results, ecosystem service supplies and demands accounting for spatial and topological patch characteristics were attributed to 14 land cover classes. The quantification of supply and demand values resulted in a positive ecosystem service budget for Zürich. The spatially adjusted service budgets and the original budgets are similar from a landscape perspective but deviate up to 50% on the patch level. The introduction of spatial and topological patch characteristics gives a more accurate impression of ecosystem service supply and demands and their distributions, thus enabling more detailed analyses in complex urban surroundings. The method and underlying data are considered suitable for urban land cover and ecosystem service mapping and the introduction of spatial aspects into relative ecosystem service valuation concepts is believed to add another important aspect in currently existing approaches.

1. Introduction

As the world population steadily grows, so do urban areas. In mid-2015 there were 7.3 billion human beings on our globe and a steady increase by at least one billion people within the next 15 years is expected, reaching 8.5 billion in 2030, 9.7 billion in 2050 and 11.2 billion by 2100 (United Nations, 2015). During the past decades, the world's urban population has grown rapidly from 746 million in 1950 to 3.9 billion in 2014 and by 2050, it is expected that 66% of the world's population will be living in urban areas (United Nations, 2014). This raises questions of how anthropogenic influence might exert stress on urban ecosystems and calls for sustainable urban development, e.g. through expansion of urban green infrastructure for the preservation and

improvement of living quality in urban areas, not only in Asia and Africa where urbanization is believed to predominately occur, but also especially in Europe under current land use change trends (Maes et al., 2015).

One indicator to express ecological functionality and its implications for humans are ecosystem services. Having emerged from the field of conservation biology, the concept of ecosystem functions and services (Daily, 1997; Millennium Ecosystem Assessment, 2005) and their valuation (Costanza et al., 1997; de Groot et al., 2002, 2012) has been widely discussed and continuously extended over the past decades. The predominant practice of attributing a monetary value in form of benefit transfers to the presence of ecosystems is however considered problematic for several reasons (Davidson, 2013). One major shortcoming addressed in this study is the general disregard of spatial patch

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characteristics and their influence on ecosystem services although many studies emphasize the importance of integrating spatial attributes in ecosystem service analyses (Alberti, 2005; Syrbe and Walz, 2012; Bagstad et al., 2013; Turner et al., 2013). Recent developments in ecosystem service studies evaluate services in terms of synergies, trade-offs and bundles (Raudsepp-Hearne et al., 2010; Larterra et al., 2012; Maes et al., 2012; Martín-López et al., 2012) and introduce a relative rather than an absolute valuation approach. One of these approaches where there is a huge potential for remote sensing data to contribute to is through land use and land cover (LULC) evaluation in terms of both potentials to provide and support ecosystem services but also of the lack and demand for ecosystem services (Burkhard et al., 2012). Another problem in adequate ecosystem service valuation arises when urban ecosystem services are considered. Problematic here are amongst others the highly subjective valuation of social ecosystem services in ever more culturally diverse societies, competing beneficiary groups in limited urban space and the fact that the effects of anthropogenic activities upon underlying ecological functions are not yet fully understood.

Despite these limitations and with rapidly expanding urban areas in mind, remote sensing can contribute considerably in deriving urban LULC where no other data is available or where in-situ data collection is problematic and resource-intensive, e.g. in areas subject to uncontrolled urban sprawl. Remote sensing can be useful for assessments of ecosystem services in several ways, either through the direct derivation of biophysical and topographic parameters that relate to ecosystem functions or indirect through the derivation of relevant LULC information that serves as proxy for ecosystem services. Several reviews highlight and summarize the contribution, potential and suitability of integrating remote sensing data into ecosystem service studies (Feng et al., 2010; Ayanu et al., 2012; Andrew et al., 2014; de Araujo Barbosa et al., 2015). However, remote sensing of urban ecosystem services is still a very new field that is not covered and only marginally mentioned in these reviews. Studies that derive detailed ecosystem service relevant information with remote sensing over urban areas are scarce (Mathieu et al., 2007a, 2007b; Lakes and Kim, 2012; Haas et al., 2014) and generally lack the integration of spatio-temporal components or only target particular services or functions.

ESA's Sentinel missions share the goal of providing us with reliable and timely information on land, ocean and atmosphere for energetic, climatological and security related topics (Aschbacher and Milagro-Pérez, 2012) and are expected to provide us with a global view of environmental parameters with high spatial and temporal resolutions crucial for natural resource management and climate/environmental research (Berger et al., 2012). Malenovsky et al. (2012) discuss and categorize the potential use of Sentinel missions for scientific observations and application domains and Sentinel-1/2 data is often given high priority for the derivation of biotic and abiotic land surface variables that constitute and define ecosystem functions, e.g. biodiversity, biomass, land surface temperature, vegetation and related indices, topography or soil types or surface soil moisture. Although the potential of Sentinel missions has been highlighted in works referred to above, effective methods to fuse, and analyse data need to be developed and the anticipated applicability tested.

Land cover mapping in complex urban environments is challenging (Ban et al., 2010) as the mixture of natural and synthetic objects and their functionalities are challenging to separate. Previous efforts related to urban land cover mapping with respect to ecological relevant space are traditionally based on medium-resolution multispectral data (Kreuter et al., 2001; Jenerette et al., 2011; He et al., 2017). Even some studies using high-resolution multispectral data exist (Lakes and Kim, 2012; Mathieu et al., 2007a, 2007b; Qian et al., 2015a, 2015b) and further studies using high-resolution data are anticipated. Ban et al. (2014) review the fusion of SAR and multispectral data for urban land cover mapping and change detection where state of the art fusion and change detection methods are presented and where the fusion potential

is highlighted. Among the few studies using Sentinel-1/2 data, Ban et al. (2017) used fused data to successfully extract urban areas and small towns in ten geographically diverse areas around the globe. Pesaresi et al. (2016) investigated the complementarity of Sentinel-1 and Sentinel-2 data for improving global human settlement mapping. The results showed that noticeable improvements in classification quality could be obtained from amongst others the simultaneous use of Sentinel-1/2 data. Yesou et al. (2016) deemed the combined use of Sentinel-1/2 data as a powerful data source for monitoring rich, complex and sensitive large ecosystems in based on a wetland monitoring study. Chang and Shoshany (2016) investigated the synergetic use of Sentinel-1/2 data for Mediterranean shrublands biomass estimation and found the fusion model generally resulting in more accurate biomass estimations as compared to using only one of the sensors. However, no fusion study for ecosystem service mapping in general, least in urban areas could be found.

One aspect and objective of this study is to evaluate a Sentinel-1 C-band SAR satellite and a simulated Sentinel-2 multispectral image product regarding their combined suitability for the classification of ecologically important urban and peri-urban space. Thus, a method is established to utilize fused Sentinel-1/2 data to derive urban and peri-urban LULC for ecosystem service supply and demand mapping at the example of the Zürich metropolitan area. Further, the ecosystem budget concept (Burkhard et al., 2012) was applied and extended by integration of often neglected spatial factors into estimation of ecosystem service supply and demand.

2. Study area and data

To exemplify the proposed method, Zürich as Switzerland's largest city with an increasing urban population of over 400,000 inhabitants and about 1.8 million people residing in the metropolitan area was chosen. Zürich's living quality is considered high with noticeable progress in terms of environmental quality over the past two decades. Major urban classes are continuous and discontinuous urban fabric, industrial/commercial areas, infrastructure in terms of the road/rail-road network including Zürich airport, construction sites, green urban spaces, sports/leisure facilities and allotments. The urban hinterland is characterized by Lake Zürich, agricultural land and forest. Agriculture is predominately defined through crops (51%), pasture (24%) and natural grasslands (20%). Vineyards and orchards play a subordinate role. The study area comprises 352 km² (16 by 22 km) and includes Zürich, the city's surroundings and the northern part of Lake Zürich as displayed in Fig. 1.

Two images were used in the study. The first one is a Sentinel-2A scene, simulated with high-resolution airborne imaging spectrometer (APEX) data. The APEX image is nadir-normalized, spectrally re-sampled, geo-corrected and orthorectified to the Swiss National Grid (CH1903). The scene dates from 2011-06-26 and includes all spatial and spectral characteristics corresponding to a Sentinel-2 level 1c product, i.e. four 10 m resolution bands (2, 3, 4 and 8a), six 20 m resolution bands (5, 6, 7, 8b, 11 and 12) and three 60 m resolution bands (1, 9 and 10). Table 1 below depicts all bands, their corresponding bandwidths and resolutions. Laurent et al. (2014) provide an extensive description of the simulated scene.

The second image that was used is a Sentinel-1A C-band SAR Interferometric Wide Swath (IW) scene in dual polarisation mode (VV + VH) dating from 2015-03-16. IW-scenes have a swath of 250 km at 5 × 20 m resolutions and C-band SAR operates at 5.405 GHz. The Sentinel-1A satellite circulates sun-synchronously on a polar orbit at an altitude of 693 km. This Level-1 GRD product consists of focused data that has been detected, multi-looked and projected to ground range using an Earth ellipsoid model. The typical product size is 1 GB (in IW mode). The ellipsoid was corrected using the terrain height specified in the product general annotation. Fig. 1 shows the Sentinel-2 image to the left and the Sentinel-1 image to the right.

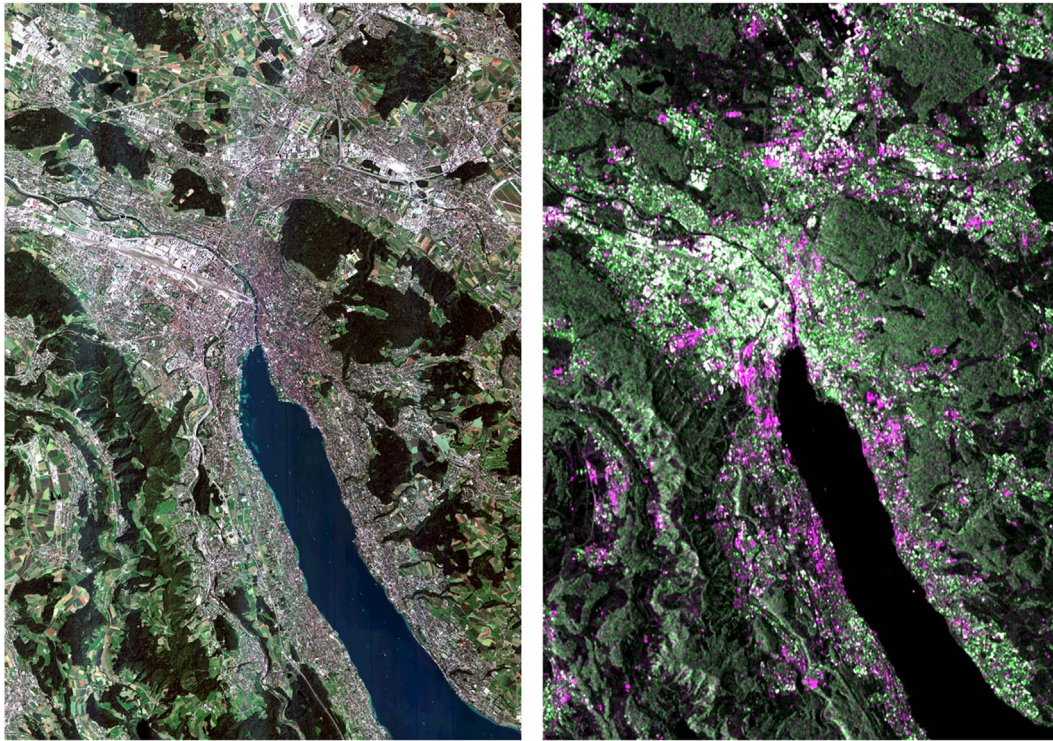


Fig. 1. RGB true-colour-composite of the Sentinel-2A APEX scene from June 2011 (left) and Sentinel-1A 5×5 adaptive Lee speckle filtered intensity data from March 2015 (VV-VH-VV) (right).

Table 1
Overview of all 13 Sentinel-2 bands, their central wavelengths and spatial resolutions.

Bands	Central wavelength (μm)	Resolution (m)
Band 1 – Coastal aerosol	0.443	60
Band 2 – Blue	0.490	10
Band 3 – Green	0.560	10
Band 4 – Red	0.665	10
Band 5 – Vegetation Red Edge	0.705	20
Band 6 – Vegetation Red Edge	0.740	20
Band 7 – Vegetation Red Edge	0.783	20
Band 8b – NIR	0.842	10
Band 8a – Vegetation Red Edge	0.865	20
Band 9 – Water vapour	0.945	60
Band 10 – SWIR/Cirrus	1.375	60
Band 11 – SWIR	1.610	20
Band 12 – SWIR	2.190	20

3. Methods

The main methodological steps involve image pre-processing and co-registration, image segmentation and classification, accuracy assessment, post-classification, spatial patch analysis and ecosystem service budget modeling as presented in Fig. 2.

3.1. Pre-processing and co-registration

The Sentinel-1 SAR image was first geometrically corrected through the Range-Doppler Terrain Correction by the method of Small and Schubert (2008) with a 3 s SRTM elevation model. Inherent speckle in SAR images is caused by the coherent interference of waves reflected from many elementary scatterers and complicates image interpretation by reducing the accuracy of image segmentation and classification (Lee et al., 1999). The reduction of speckle is therefore performed in this study by application of a 5×5 adaptive Lee filter that, empirically compared to kernel sizes from 3×3 to 11×11 and Frost and Gamma Map filters, yielded the best results. The applied filter with a 5×5 kernel

size has also been found suitable in other studies (Lopes et al., 1990; Mansourpour et al., 2006).

The Sentinel-1 SAR image was then co-registered to the Sentinel-2 MSI image with RMSEs less than 0.5 pixels in X and Y directions. The filtered Sentinel-1 bands and the Sentinel-2 RGB, NIR and two SWIR bands were resampled to a 10 m spatial and to 8-bit radiometric resolution with a 2% tail trim.

3.2. Image segmentation

Image segmentation was performed with KTH-SEG, an edge-aware region growing and merging algorithm (Ban and Jacob, 2013). By creating an edge/no-edge decision layer using an enhanced Canny edge detector, segment growing is divided off-edges and along edges. The homogeneity criteria for both growing and merging are defined by a weighted sum of change in mean and change in standard deviation. Merging is performed using a mutual best neighbour approach, followed by threshold merging. Growing is limited to the minimum segment size and merging to the maximum segment size. For the segmentations in this study, the following parameters were empirically determined and were found to generate the most suitable result: Canny threshold: 0.1–0.2; grow 0.5/0.5, merge 0.5/0.5, minimum and maximum segment sizes were chosen from 5 and 100 pixels.

3.3. Image classification

The following classes originating from the CORINE land cover classification scheme (Bossard et al., 2000) were chosen according to the ecosystem service supply and demand budgeting scheme used by Burkhard et al. (2012): Continuous urban fabric; discontinuous urban fabric; industrial or commercial units; road and rail networks and associated land; airports; construction sites; green urban areas; sport and leisure facilities; complex cultivation patterns (allotments); agriculture and natural vegetation; mixed forest; water courses and water bodies. Vineyards and fruit trees/berries that are also present in the study area were excluded from the classification approach since they play a minor

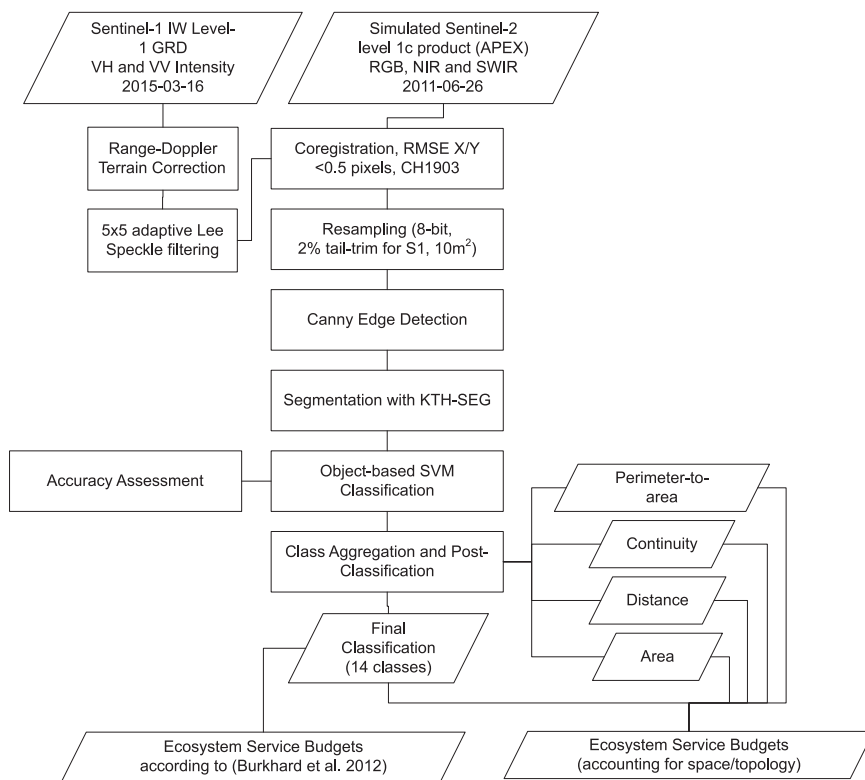


Fig. 2. Methodology flowchart.

role and would rather contribute to higher class confusions than to adding a meaningful ecosystem service class. Vineyards and fruit trees/berries are hence included in the agriculture and natural vegetation class.

After edge detection and image segmentation, the segment type was determined through object-based Support Vector Machine (SVM) classification (Chang and Lin, 2011; Cortes and Vapnik, 1995) into initially 19 spectrally different classes. These classes were then aggregated into 12 contextually similar information classes. SVM is considered to be an effective and popular machine learning classifier that has proven valuable in classification of remotely sensed data and that is suitable for object based image analysis (OBIA) approaches (Blaschke, 2010).

3.4. Accuracy assessment

Classification accuracy assessment was performed by selecting a minimum of 1000 validation pixels homogeneously distributed across the images for each class. Validation site selection was performed under two premises, i.e. that all instances of a class were covered in an appropriate amount and that the areas were equally split over the entire study area. Accuracy assessment was performed directly on the classification result prior to post-classification modifications through sieving and reclassification. As measures of accuracy, kappa coefficient and overall, average, user's and producer's accuracies were chosen.

3.5. Post-classification

The classification result was then sieved (filtered) to remove unwanted pixel aggregations of no more than 300 m² through merging with largest neighbouring segments. These spurious elements are considered as misclassifications will otherwise erroneously be treated as patches in the landscape metric analysis and confound the result. Finally, urban green spaces and urban forests were determined through reclassification, i.e. sport and leisure facilities and agriculture/natural vegetation were reclassified into urban green spaces and mixed forest was reclassified into urban forest under an urban mask.

3.6. Spatial patch analysis

From the final classification, the four spatial characteristics of land cover patches that are believed to influence ecosystem service provision capacities and demands were derived under these following four assumptions:

3.6.1. Distance

It is assumed that natural land cover patches that provide ecosystem services are more valuable to urban dwellers if they are easier accessible. Therefore, a Euclidean distance map from the boundary of urban, man-made classes into natural land cover patches was generated. The smaller the distance, i.e. the closer the natural land cover patches are to the edge of urban areas, the higher their value. Pixel values decrease continuously with increasing distance from urban areas. Vice versa, a distance map was created from the boundary of man-made structures and natural land cover classes away from the boundary towards to man-made patch centre. The further away the cell is located from the boundary, the higher the demand. For each patch, the distance values were averaged and the distance mean attributed to the patch.

3.6.2. Perimeter-to-area-ratio

The perimeter-to-area ratio expresses shape complexity. The reason for choosing that metric is the assumption that natural areas furthest away from edges shared with other classes are least affected by other natural or anthropogenic influences (e.g. through sound or air pollution of adjacent roads), thus being more pristine and most likely to provide the services initially ascribed to the respective class. Hence, the lower the ratio, the simpler the patch shape, the less edge shared with surrounding patches and thus the higher the service provision. Perimeter-to-area ratios were calculated on patch as opposed to on landscape level. Regarding artificial patches, a more complex structure is desirable since more edge can be shared with natural land cover. Simpler structures with the central patch area far away from ecosystem service providing areas are considered of being in higher service demand.

3.6.3. Area

This factor originates from initial valuation approaches, e.g. Costanza et al. (1997) or de Groot et al. (2012) where total ecosystem values are determined by multiplication of a fixed pre-defined class service value with the total area of that particular class in a landscape. In Burkhard et al. (2012), the relative valuation scheme does not regard areal influence on service provision or demand but the authors believe that it should be considered. Larger patches of the same class are likely to offer higher species richness and diversity and are considered to be of greater recreational value to the urban population. The area in hectare was calculated for all patches that were then ranked from the largest patch being attributed highest score to the lowest patch for provisional patches. The same relation holds true for patches that lack services. Larger artificial patches with the patch centre far away from ecosystem service provisional areas are negatively influenced by surrounding artificial structures, e.g. in terms of microclimate or lacking water infiltration capacities, thus contributing to increased temperatures and increased flood risks in central urban areas. These artificial patch cores are thus considered of being in higher service demand. For each patch, the distance of each cell area was summed up and averaged over the polygon.

3.6.4. Contiguity

High patch connectivity and less fragmented landscapes are considered beneficent in several ways, e.g. for species dispersal or recreational purposes. Contiguity values were calculated for all land cover patches and the best-connected patches were awarded highest contiguity scores. Highly connected urban areas are considered to be in higher demand of services since regulating service-provisioning areas can be interrupted and degraded through a highly connected artificial network. Therefore, artificial patches with high contiguity are considered being in higher demand.

In this study, land cover at the patch level as opposed to a landscape centred perspective is investigated since the topological patch distributions and the spatial characteristics of individual patches are believed to be most relevant factors for the evaluation of ecosystem budgets. If inter-regional or multi-temporal comparisons are aspired, characteristics of the landscape as a whole might even reveal diverse budget patterns and different development trajectories.

Eight spatial influence maps were generated in total that modify service provision (four maps) and service demand (four maps). Each influence map was linearly scaled so that the lowest value for each spatial characteristic was assigned 0.5 whereas the highest value was set to 1.5. A value of 1 describes the average patch condition in the landscape that would not exert any influence on ecosystem services. Thus, spatial patch characteristics are allowed to alter ecosystem service supply and demand not more than to 50% of their initially suggested budget.

3.7. Ecosystem service budget modeling

As a final step, two ecosystem service budget maps were derived. The first map was created by attributing the original ecosystem service demand and supply scores for 22 regulating, provisioning and cultural ecosystem services and 7 ecological integrity measures as presented in (Burkhard et al., 2012) and in Table 2 to each land cover class.

The supply values attributed to each class are defined as the sum of all ecological integrity, regulating, provisioning and cultural services and mirror the capacities of ecosystems and their functions to supply services. Areas that lack the provision of these services are considered neutral or being in service demand based on human interaction, their structural design, use and functioning. A second supply and demand map was generated with modified service values through addition and averaging of all influence maps and through multiplication with the original budget values. In order to enable inter-urban comparisons, supply and demand budgets were related to the LULC in the study area

by multiplying the area with the attributed budget values per class. Based on the difficulty in putting price tags on ecosystem services, the fact that there is until today no homogeneous valuation scheme for ecosystem services in urban areas and following the trend of rather analysing ecosystem demands and services (Burkhard et al., 2012; Kroll et al., 2012), no absolute monetary changes in ecosystem service are considered here. Instead, the budgets are expressed as service units per hectare.

4. Results

In total, 5339 segments distributed over 19 spectrally different classes were generated by KTH-SEG that were then classified and aggregated into 12 contextual classes. Table 3 below shows the classification confusion matrix. The classification into urban and peri-urban classes resulted in an overall accuracy of 79.81% with a Kappa coefficient of 0.78. Largest confusions occurred between construction sites and industrial/commercial areas and between the built-up classes of discontinuous urban fabric, sport and leisure facilities and between airport runways and roads.

Fig. 3 below shows the classification result after post-classification and sieving.

The distinction of allotment gardens was not intended in the beginning of the study since their classification is considered difficult because of the class's multimodal spectral reflectance and backscatter patterns due to the different land cover types constituting allotment gardens. Here, the combined use of SAR and MSI data is believed to have contributed considerably by providing a unique combination of a digital number value pair to the SVM classifier. Sports and leisure facilities occur predominately in the form of tennis courts that can be distinguished through their low backscatter and distinct spectral response. Roads that are usually difficult to classify by SAR data alone due to their low backscatter profile that they share with other flat surfaces, such as bare fields, lawns or water bodies could be classified with 80% accuracy through the integration of multispectral data. Roads are difficult to distinguish by multispectral data alone since other sealed surfaces such as backyards, parking lots or airport runways and roofs especially in industrial and commercial areas show similar spectral responses. Difficulties remain in the distinction of airports that share the same spectral response in MSI data and backscatter in SAR imagery as roads and in the distinction of commercial/industrial areas, partly because these areas should be contextually separated and partly because they are more distinguished through their function (information class) than through their backscatter/spectral response that is similar to other buildings.

In the four maps presented in Fig. 4, green indicates that the spatial factor under consideration yields a positive, increasing effect on service provision, whereas red indicates negative spatial effects, i.e. less ecosystem service supplies. In the maps showing the spatial influence on service demand as shown in Fig. 5, green indicates reduced demands whereas red expresses an increased demand for ecosystem services because of detrimental spatial characteristics. Yellow in all maps expresses that no qualitative influence on the land cover patch was made indicating the average condition of a patch in the landscape.

The combined topological and spatial influence on supply and demand of ecosystem services is shown in Fig. 6 below. In the map to the left showing the combined influence on supplies, green indicates overall positively affected supply patches, yellow indicates patches that are not particularly affected by spatial patch attributes or where positive and negative influences cancel each other out. Red expresses degraded values per patch provision. In the combined service demand factor map, green indicates patches with less demand than average and red patches indicate areas where spatial patch characteristics increase service demand.

The most prominent patch in the landscape where spatial influences have a positive effect on service provision is Lake Zürich. Smaller

Table 2

Ecosystem service demand and supply scores according to and reproduced from page 24, Fig. 4 in Burkhard et al. (2012). The table is the result of a subtraction of two tables. Table B that expresses the demand for ecosystem services was subtracted from Table A that contains ecosystem capacities. The values in the table range from -5 to 5 where -5 denotes that the CORINE class is in highest demand of ecosystem services and 5 means that the class in question has a high capacity to provide ecosystem services. The column to the right is the sum of all ecosystem service scores that was used in the further analysis regarding value adaption in terms of distance, perimeter-to-area-ratio, area and contiguity. The class descriptors are as follows: A1 = Continuous urban fabric; A2 = Discontinuous urban fabric; A3 = Industrial or commercial units; A4 = Road and railway networks; A5 = Airports; A6 = Construction sites; A7 = Green urban areas; A8 = Sport and leisure facilities; A9 = Complex cultivation patterns; A10 = Agriculture and natural vegetation; A11 = Mixed forest; A12 = Water courses; A13 = Water bodies.

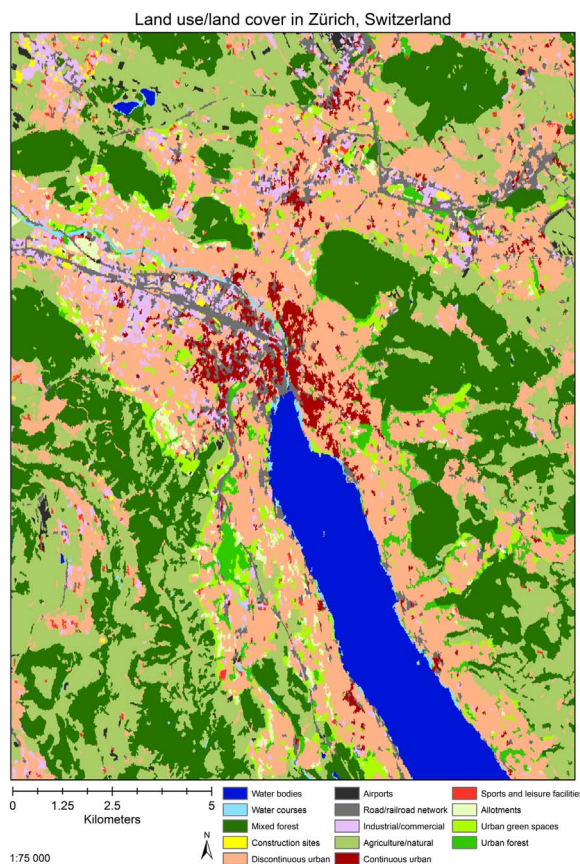
	Abiotic heterogeneity	Biodiversity	Crops	Livestock	Fodder	Capture Fisheries	Aquaculture	Wild Foods	Timber	Wood Fuel	Energy	Biochemicals and Medicine	Freshwater	Recreation & Aesthetic Values	Intrinsic Value of Biodiversity	SUM
A1	0	0	0	0	0	0	0	0	-5	-3	-4	-5	-5	-5	-1	-1
A2	1	1	0	1	1	1	1	1	-5	-3	-5	-5	-5	-5	-1	-2
A3	1	1	0	0	0	0	0	0	-1	-5	-4	-4	-5	-5	-1	-3
A4	2	2	0	0	0	0	0	0	-2	-4	-4	-4	-1	-4	-3	0
A5	1	1	1	1	1	1	2	0	-2	-5	-1	-1	-1	-4	-1	-1
A6	2	1	0	0	0	0	0	0	-2	0	-2	0	0	-1	-2	-2
A7	3	3	2	1	4	4	3	2	0	1	0	1	1	0	2	1
A8	2	2	2	1	4	4	3	2	-1	1	0	0	0	-2	1	1
A9	4	3	3	2	4	4	1	3	1	0	0	1	0	-1	1	-5
A10	3	3	3	2	3	3	2	3	2	0	1	1	1	0	2	-3
A11	3	5	5	4	5	5	5	5	5	4	3	2	2	5	5	5
A12	4	4	0	3	3	3	3	1	1	0	2	1	1	0	0	3
A13	4	4	0	4	4	4	3	4	2	1	1	2	2	0	0	1

	Water purification	Pollination	Crops	Livestock	Fodder	Capture Fisheries	Aquaculture	Wild Foods	Timber	Wood Fuel	Energy	Biochemicals and Medicine	Freshwater	Recreation & Aesthetic Values	Intrinsic Value of Biodiversity	SUM
A1	-1	-3	-5	-5	-1	-5	-5	-5	-3	-2	-4	-5	-5	-4	-2	-79
A2	-2	-4	-3	-4	-1	-4	-4	-3	-3	-3	-3	-5	-5	-4	-3	-70
A3	-3	-4	-4	-5	-4	-4	-4	-3	-5	-5	-4	-5	-5	-1	-1	-79
A4	0	-1	0	0	0	0	0	0	-2	0	-4	0	-1	-2	0	-24
A5	-2	0	-2	-2	1	-1	-1	-1	-1	0	-5	-1	-3	-1	0	-27
A6	-2	0	0	0	0	0	0	0	-4	0	-4	0	-2	0	0	-18
A7	1	-1	-1	-1	0	0	0	1	0	1	-1	0	-2	-1	-1	18
A8	0	1	-2	-2	-1	-2	-2	-2	-1	-1	-3	-3	-3	2	0	-3
A9	-2	-3	3	0	3	0	0	0	0	0	-1	1	-1	2	0	16
A10	-1	-2	2	3	2	0	0	3	3	3	0	0	-2	2	3	38
A11	5	5	0	0	1	0	0	4	4	4	1	5	0	5	5	100
A12	3	0	0	0	0	2	0	4	0	0	3	0	5	5	5	52
A13	0	0	0	0	0	2	0	4	0	0	0	0	5	5	4	50

Table 3

Confusion matrix (B1 = Water bodies; B2 = Water courses; B3 = Mixed forest; B4 = Construction sites; B5 = Discontinuous urban; B6 = Airport; B7 = Road/railroad network; B8 Industrial/commercial; B9 = Agriculture and natural vegetation; B10 = Continuous urban; B11 = Sport/leisure facilities; B12 = Allotment gardens).

	B1	B2	B3	B4	B5	B6	B7	B8	B9	B10	B11	B12	UA
B1	100	0	0	0	0	0	0	0	0	0	0	0	100
B2	0.6	91.8	3.6	0	1.5	0	0.6	0.1	1.8	0	0.1	0	91.7
B3	0	0	95.4	0	0	0	0	0	4.6	0	0	0	95.4
B4	0	0	0	79.9	0.3	0	1.2	14.1	0	0	0.4	4.1	79.9
B5	0	0	1.2	0.7	81	0	1.9	4	0.5	4.1	1	5.5	81.1
B6	0	0	0	0	0	72.4	0.7	12.8	13.5	0	0.6	0.1	72.4
B7	0	0	0	0	6.4	0.1	86.9	1.7	2.1	0	0.3	2.5	87
B8	0	0	0	39.3	0.8	2.1	10.7	31.8	4	5.2	5.8	0.4	31.8
B9	0	0	0	0	1.1	2.4	0	0	95.1	0	0	1.4	95.1
B10	0	0	1.8	0	12.1	0	5.1	1	0	80	0	0	80
B11	0	0	0	3.7	13.8	0.5	0.9	1.6	14.8	0.5	51.1	13.2	51.1
B12	0	0	0	0	13	0	0	0	6.6	0	0	80.4	80.4
PA	99.4	100	93.8	65.3	66.1	93.8	81.2	47.4	68.7	89.4	79.7	77.9	

**Fig. 3.** Classification result after post-classification and sieving.

patches within the urban boundary are negatively affected by spatial characteristics, partly because of size and edge contamination through artificial land cover classes and partly through lower contiguity values as result of fragmentation. Looking at the combined ecosystem service demands, large urban patches are most negatively affected by spatial characteristics through large distances of centrally located built-up areas to ecosystem service providing areas. Smaller artificial patches in the urban periphery are in less demand due to their proximity to green and blue spaces. Interestingly, smaller patches in the urban core south on the Limmat river bank seem to be in less service demand as opposed to patches on the northern bank, most likely through the proximity to the river and urban green spaces.

A budget map based on the original supply and demand values according to Burkhard et al. (2012) is shown in Fig. 7 to the left. In the

same figure, the spatially adjusted budget map is shown to the right.

As can be seen from Fig. 7, main areas that supply the local population with ecosystem services are urban and peri-urban forests, agricultural areas and Lake Zürich. Forests are the most important service provider due to the plethora of ecosystem functions they offer. Continuous and discontinuous urban areas and industrial/commercial areas are in highest demand of ecosystem services, obviously through the presence of humans as potential and actual beneficiaries but also partly because artificial structure such as roads and buildings do not generate services, consume energy and/or even pollute the environment. The visual and numerical differences between the original and spatially adjusted budgets are subtle as the degree of spatial influence has been limited in this study. The range of adjusted service budgets is higher through the influence of very dominant landscape patches but investigations of the frequency distribution, supply and demand values are found to be generally less extreme as can be seen from Table 4. A similar observation could be made in the study of Frank et al. (2012) who introduced a conceptual approach for how to enhance the assessment of ecosystem services with regard to landscape structural aspects. One of the findings in the study was that without the integration of spatial attributes, the actual ecosystem service provision potential of the landscape under consideration would be overestimated.

In order to gain an impression of the overall quantitative budgets in the study areas both the original and adjusted values were multiplied with total area in hectare of the corresponding land cover class. Table 5 below summarizes the classes, their respective extent in hectare, the attributed budget value from Burkhard et al. (2012) and the spatially adjusted budget values generated in this study.

The overall budgets for the study area were calculated by summation of all hectare-units. With the original values, the total original service budget amounts to a supply surplus of 446,227 (1160,088 supply-units – 713,861 demand-units) units as opposed to 446,707 (1031,695 – 584,988) calculated with the adjusted budget.

As can be seen in the final budget maps in Fig. 7, mixed forest is the by far largest contributor to service supplies, not only through the attributed high supply values but also through the large forested area in the landscape. Agriculture and natural vegetation also share close to a fourth of the total landscape area but their provision of function and services is considerably lower than the ones for forest. Discontinuous urban areas cover the largest part of the landscape and account for about 77% of the total demands in the area. The per class deviations through spatial adjustments range from 25% for urban forest to 2% for water bodies.

Summation of the budget values resulted in the same total service supplies for the original and the newly developed approach but investigations of individual patch values show differences up to 50% that can give a better indication of relative importance of landscape patches.

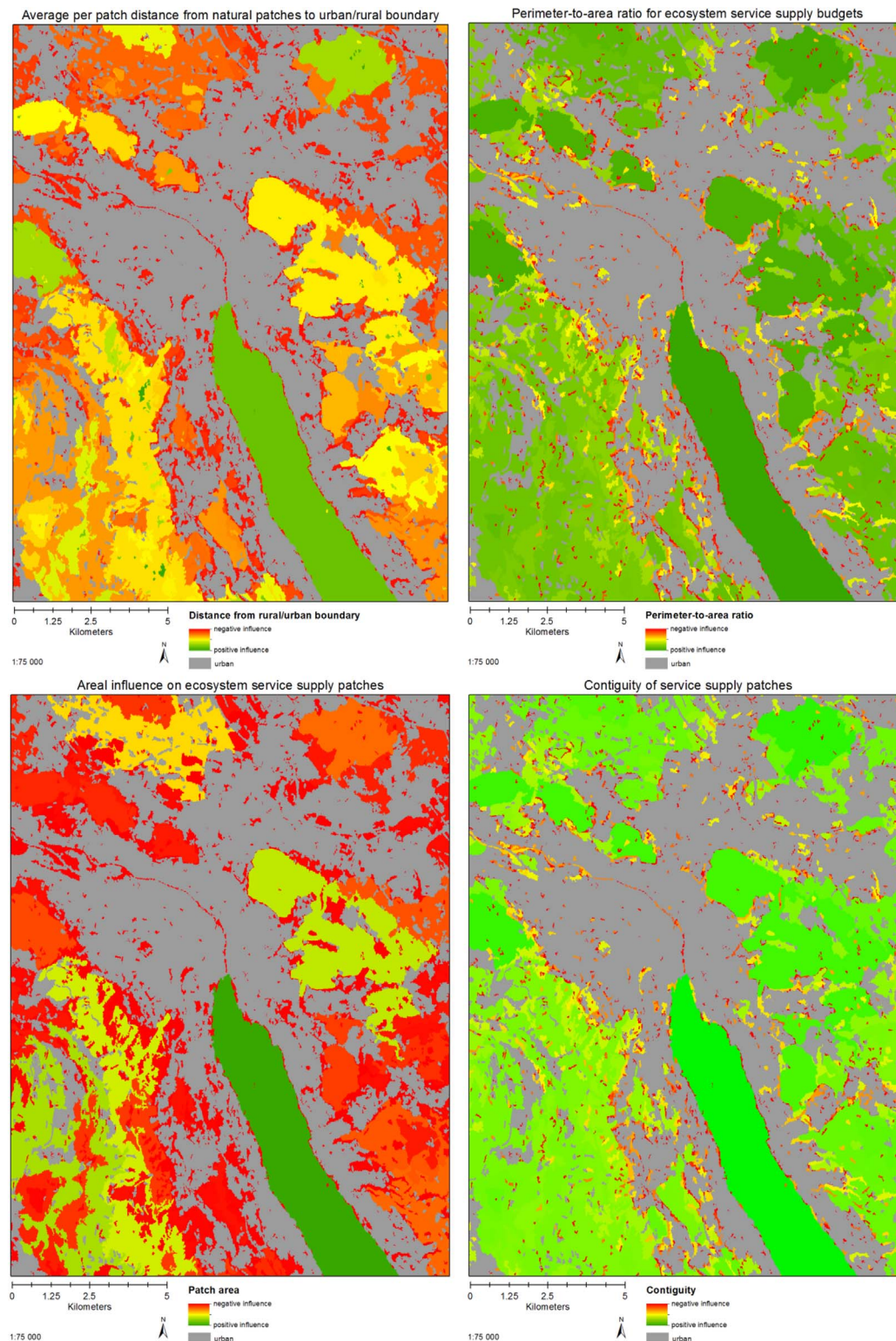


Fig. 4. Influence of the four spatial characteristics on service supply patches. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

5. Discussion

This study demonstrated the potential of Sentinel-1A SAR and Sentinel-2A MSI data for urban land cover mapping. The approach of using fused data is believed to be an advantage in the classification strategy but class confusions remain. One reason for this is that the relevant classes according to CORINE scheme do not only represent land cover but also land use. As it is easier to distinguish land cover through different spectral responses and backscatter profiles, accurate

land use mapping is still very challenging using remote sensing data alone, particularly in complex urban areas where the same land cover class, e.g. built-up is attributed different functionality expressed in land use categories, e.g. residential, commercial, industrial or infra-structural. As the current classification strategy is based on brightness values and standard deviations, no topological or shape attributes of the segments are drawn into consideration in the SVM classifier. These could be integrated in the future to reduce class confusion. Image segmentation was computationally speaking more complex than SVM

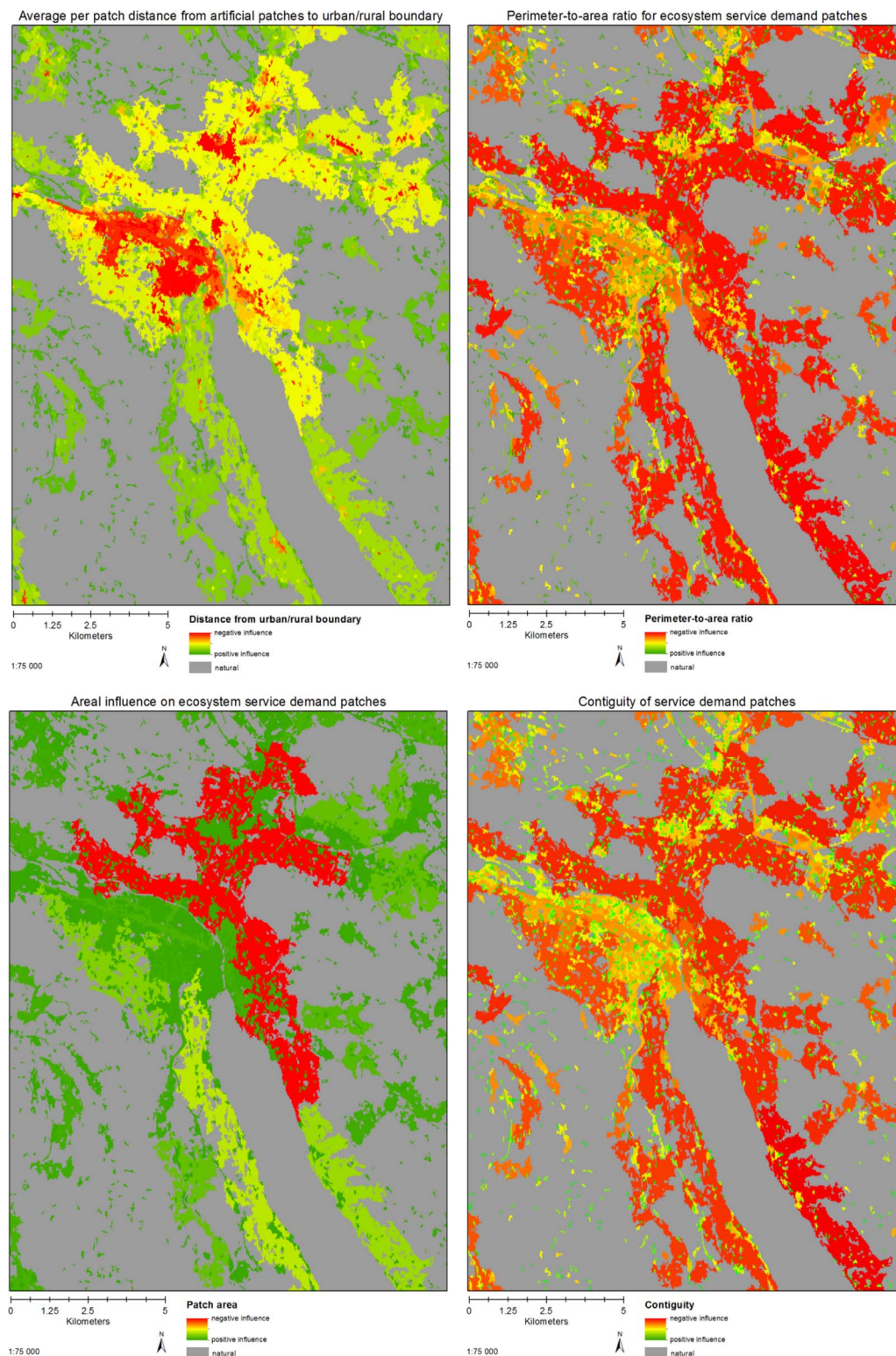


Fig. 5. The influence of the four spatial characteristics on service demand patches. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

classification and was therefore far more time consuming. The classification into 19 spectrally different classes only took a couple of seconds to perform.

It could be shown that the suggested classification approach worked well on fused MSI and SAR data (ca. 80% classification accuracy). Well known misclassifications with radar (the distinction of different classes with similar surface backscatter patterns, i.e. roads, runways, still water or lawns) and with multispectral data (the distinction of classes with similar spectral reflectance, e.g. bare soil/construction sites/built-up

space or dark green vegetation in forms of forest and agriculture) could partly be reduced by the fused approach. The potential of fusing radar and optical data has been demonstrated before (Ban et al., 2010; Ban and Jacob, 2013) and could be confirmed by this study. The unique combination of spectral response from the Sentinel-2 MSI data and backscatter from the SAR instrument at C-band resulted in the initially not planned allotment garden/complex cultivation pattern class. Apparently, the segments that were created in the segmentation process based on those brightness values and standard deviations of all image

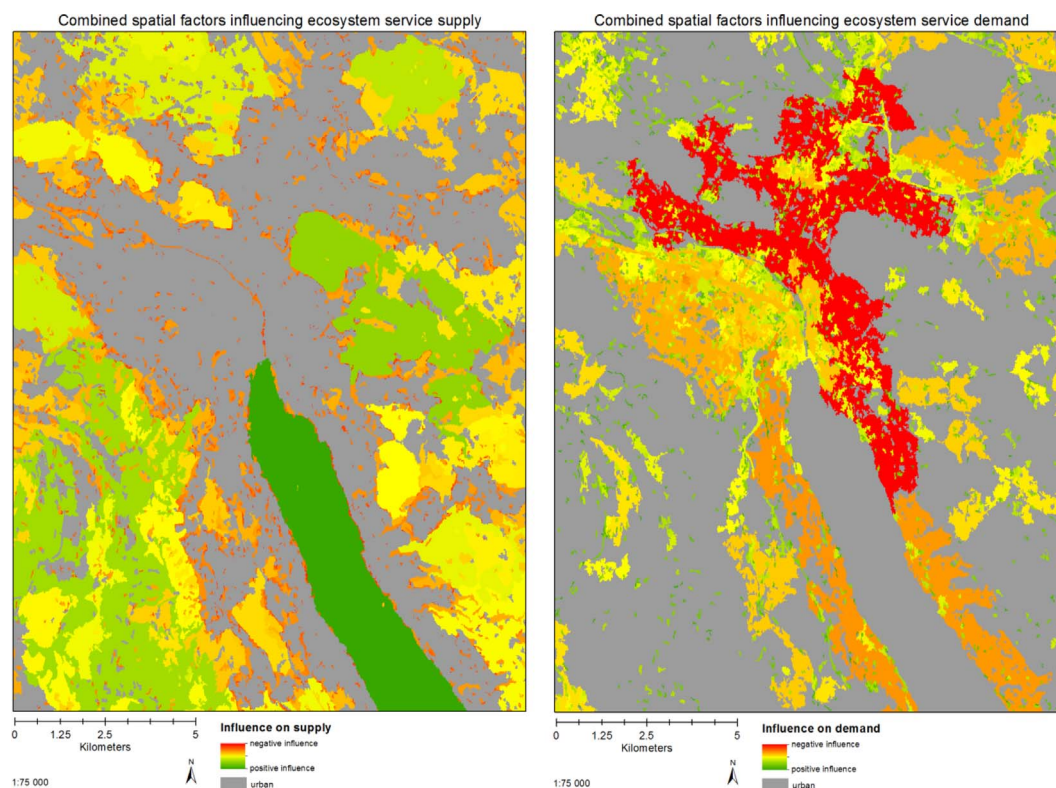


Fig. 6. Combined influence of the four spatial patch characteristics on provisional supply patches (left) and on demand patches (right). (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

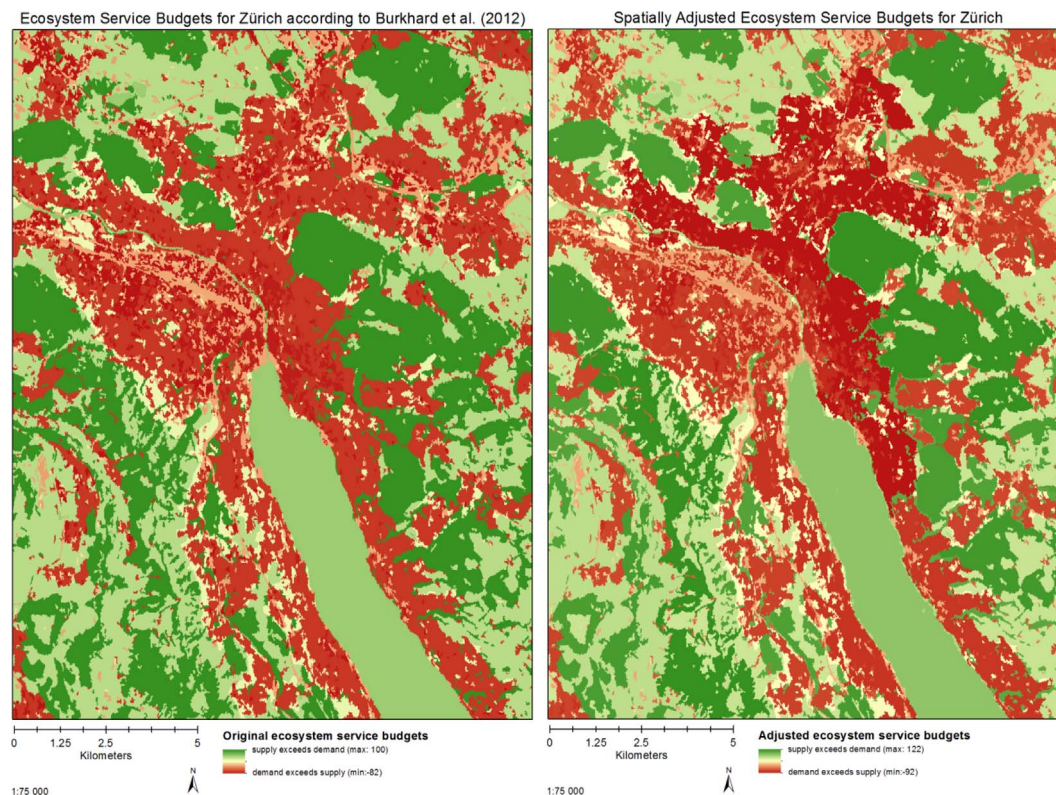


Fig. 7. Ecosystem service budgets according to Burkhard et al. (2012) (left) and spatially adjusted (right).

bands were unique. There might even be other classes present in the study area that are believed to be difficult to classify and that are not anticipated but that might easily be distinguished by the fusion approach. In order to detect those classes decoupled from a particular

classification scheme such as CORINE, an unsupervised classification method could be applied to the dataset. The selection and grouping of pixels or segments devoid of a priori expectations might reveal additional classes and thus imply another benefit of the combined approach.

Table 4

Class overview with original (here is referred to Table 2) and modified ecosystem service budgets, their mean, minima, maxima and standard deviations.

Class	Original balance (BV) (Σ supply $-\Sigma$ demand)	Spatially adjusted mean budgets (BV _{adj})	min	max	Std. Dev.
Continuous urban fabric	−79 (1 – 80)	−67.91	−84.2	−43.6	7.3
Discontinuous urban fabric	−70 (11 – 81)	−56.76	−90.7	−37.7	7.0
Industrial/commercial units	−82 (3 – 85)	−69.69	−91.8	−48.4	6.7
Road/rail networks/assoc. land	−23 (5 – 28)	−18.99	−25.9	−12.2	2.0
Airports	−27 (8 – 35)	−21.89	−26.3	−16.5	2.1
Construction sites	−18 (3 – 21)	−14.97	−19.6	−11.8	1.5
Sport and leisure facilities	−3 (30 – 33)	−2.65	−3.1	−2.3	0.1
Complex cultivation patterns	16 (37 – 21)	14.26	12.8	15.9	0.5
Agriculture/natural vegetation	38 (57 – 19)	35.93	29.4	45.6	2.6
Mixed forest	100 (103 – 3)	86.60	50	122.3	16.0
Water courses	52 (53 – 1)	46.65	41.6	59.6	2.9
Water bodies	50 (51 – 1)	49.05	42.4	69.2	6.4
Urban green space	18 (34 – 16)	16.18	14.3	17.9	0.7
Urban forest	100 (103 – 3)	74.80	50	101.7	15.1

Table 5

Land cover classes, their total area in hectares present in the study area, original and spatially adjusted ecosystem service supply and demand budgets.

Class	LULC in hectare	LULC in percent	Original budgets (LULC*B _V)	Spatially adjusted budgets (LULC*B _V _{adj})
Continuous urban fabric	615	2.10	−48,618	−41,793
Discontinuous urban fabric	7788	26.56	−545,229	−442,103
Industrial/commercial units	1023	3.49	−83,845	−71,258
Road and rail networks and associated land	1264	4.31	−29,082	−24,012
Airports	169	0.58	−4560	−3697
Construction sites	115	0.39	−2062	−1715
Sport and leisure facilities	155	0.53	−465	−410
Complex cultivation patterns	401	1.37	6416	5718
Agriculture/natural vegetation	6983	23.82	265,350	250,896
Mixed forest	6897	23.53	689,743	597,317
Water courses	165	0.56	8594	7709
Water bodies	2185	7.45	109,271	107,194
Urban green space	915	3.12	16,471	14,806
Urban forest	642	2.19	64,244	48,055

One advantage of the CORINE classification approach is that it is well established and not designed for a particular application domain, i.e. it can be applied for many purposes. From an ecosystem service perspective, well established class definitions based on ecosystem functions or function bundles would be an interesting contribution to avoid confusions between land cover and land use and to facilitate a clearer distinction between service provision capacities.

The ecosystem service approach presented here is an extension of the relative valuation concept expressing both supply of and demand for ecosystem services that is attributed to LULC classes. Spatial characteristics of patches are believed to influence ecosystem service supply and demands and are introduced to give more accurate and detailed information on service patches. The choice of spatial attributes was made based on selected literature that suggests either positive or negative effects of spatial attributes on service provision. The fact that spatial characteristics of patches, landscape fragmentation or landscape connectivity have an impact on biodiversity, ecological functioning and thus ecosystems is widely acknowledged. In what ways and to what degrees these impacts are to be addressed and quantified in a systematic way is however a matter of discussion.

Through the scaling approach presented here, the overall absolute spatially adjusted budgets were similar to the original ones from a landscape perspective, but supply and demand scores deviated up to 50% of their original values on the patch level. The scaling, weighting and choice of spatial metrics define the degree and type of influence on service provision and demand. In this study, the importance of the four implemented spatio-topological patch characteristics was considered

equal since this study aims at introducing the influence of spatial patch characteristics on ecosystem service provision without a particular service, class or specie in mind. These spatio-topological considerations presented here are also non-exhaustive and introduced to the ecosystem service supply and demand concept since a relative and qualitative distinction should be made between patches. Factor scales and weights are adaptable and can be changed to particular objectives, services, functions or class preferences or when further reliable couplings between ecosystem functions and spatial patch attributes have been established. It is understood that the spatial attributes of patches do not influence every underlying ecosystem function in the same way. For particular applications, e.g. the analysis of a biotope network for a particular species (habitat function), contiguity and fragmentation might be far more important than patch complexity expressed as perimeter-to-area ratio. On the other hand, for regulating services such as flood control functionality, contiguity might play a subordinate role in comparison to areal aspects of water bodies. A comprehensive framework that evaluates every ecosystem function in relation to landscape structural aspects and patch characteristics would give much more appropriate estimations of actual supplies and demands and the development of such a framework and closer links between service supply and demands and landscape composition and configuration are recommended for further research.

Through the integration of spatial factors in this new approach, not only the mere presence or absence of ecosystem services can be determined but even a distinction of ecosystem service provision or demand quality can be made, resulting in more accurate budget values.

The spatial distribution of supply and demand patches in relation to their neighbours can in conjunction with their spatial attributes help in determining supply and demand hotspots and highlight the importance of particular patches over others. The factors used in this approach can easily be modified to analyse a particular aspect of service supply or demand.

As noted before, not all ecosystem functions are influenced to the same degree by spatial and topological aspects, especially provisioning services are predominately dependent on patch size, not so much on their spatial distribution. On the other hand, especially for cultural services, accessibility and thus proximity of residential areas to providing patches is most important. This aspect can be addressed by integrating population distribution data to evaluate service demands and provision with respect to population density, since it is human beings that eventually gain advantages or disadvantages from the presence or absence of ecosystems. Densely populated areas are in higher demand of certain ecosystem services, e.g. cultural and recreational services and thus the importance of provisional patches in relation to the size of nearby benefiter groups should be recognized as well. At the same time, over-use and exploitation of services can deteriorate service provision. This can be exemplified at the example of an urban park. A park is considered more valuable, when more people enjoy its presence. At one point, the provisional capacity is however saturated and the presence of more visitors would result in service reduction in terms of the destruction of lawns, littering, vandalism, noise and air pollution, lack of space for recreational activities, entrance queues, etc. Ecosystem services are defined as benefits that humans gain from functioning ecosystems. Humans in that sense a very broad term not distinguishing between where people live or between socio-economic differences. Not all services are equally important for everyone at the same point in time and in urban areas with limited space, humans compete for resources and services. Finding new ways of adequately integrating these differences into valuation approaches would be another asset.

6. Conclusions

This study investigated the potential of fused Sentinel-1A SAR and Sentinel-2A MSI data for urban land cover mapping and ecosystem service budgeting solely relying on remotely sensed data, thus presenting a generic approach for location-independent and rapid ecosystem service assessments. Classification accuracies of about 80% suggest the suitability of fused Sentinel-1/2 data and the proposed method for urban land cover mapping and classification of ecological important space. Further use of these datasets in this application domain is recommended. Through the combined use of C-band SAR and multispectral data, the multi-feature class of allotment gardens/complex cultivation patterns could be unexpectedly well distinguished. The concept of expressing ecosystem service budgets according to Burkhard et al. (2012) was extended to include spatial properties and topological relations of landscape elements. The magnitude of influences these aspects have on service scores are limited to alter originally proposed budgets not more than 50%. The influence of spatial factors was exemplified by four landscape patch characteristics: patch size, distance to adjacent classes, perimeter-to-area ratio and contiguity. Two ecosystem service supply and demand budgets were calculated. The original budgeting scheme and the newly proposed one accounting for spatial patch characteristics resulted in similar overall budgets in Zürich, Switzerland. The new scheme that integrates spatial characteristics resulted in different, more detailed per patch service supply and demands, enabling more comprehensive investigations of particular areas that have the capacity to provide or are in demand of services. The quantification of spatially adjusted service budgets enables inter-urban and inter-regional comparative studies and aids in identifying local capacity and demand hotspots. With that information, according planning measures can be taken, e.g. expansion and preservation of main ecosystem service contributors and redesign or reduction

of demanders. The Zürich metropolitan study area showed positive ecosystem service budgets with forest, agricultural/natural areas and Lake Zürich as largest contributors to ecosystem service supplies. Discontinuous urban fabric as dominant land cover class showed to account for 77% of the total demand.

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